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Improving the hydrogen storage properties of lithium hydride (LiH) by lithium vacancy defects: Ab initio calculations

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ABSTRACT

Using the coherent potential approximation (CPA) integrated into the AkaiKKR package, the effect of lithium vacancies on the temperature of desorption and the energy of formation in the lithium hydride (LiH) is investigated. Firstly, the calculations show that the LiH has high stability and consequently a high decomposition temperature, and that is due to the strong interaction between Li and H atoms. Secondly, the results reveal that the Li vacancy defects have a beneficial effect on the hydrogen storage properties of the hydride by increasing its formation energy and gravimetric density, and decreasing its desorption temperature and stability. Two reasons can explain this decrease in stability as a function of the concentration. First, there is a decrease in Li atoms that form strong bonds with hydrogen atoms. Second, the lithium and hydrogen states shift towards the conduction band, which is shown using the density of states (DOS).

1. Introduction

The use of renewable energy sources (solar, biomass, wind, hydroelectric and geothermal energy) for electricity production faces some difficulties such as the control and management of the electrical grids. In this context, hydrogen (H_2) can be the optimal solution for addressing these challenges by efficiently converting electrical energy into a hydrogen-based form [1]. Nowadays, the compressed hydrogen storage in the gaseous form (70 MPa) is the most common method, but the safety of high-pressure hydrogen must be improved. Storing hydrogen in liquid form at a temperature of −253 °C can be an effective solution; however, this method requires significant energy and expenses for the liquefaction of hydrogen. The adsorption of hydrogen on solid materials such as carbon nanotubes and metal-organic frameworks is another solution, but the low volumetric storage capacities and the operating temperature of those materials limit their use for hydrogen storage [2–13]. Recently, more attention is given to a promising method that depends on the metal hydrides to absorb hydrogen. Our work will focus on this method with particular focus in lithium hydride (LiH), which is a promising solution for hydrogen storage applications.

The LiH figures among the lightest metal hydrides with a high gravimetric density up to 12.6 wt% and an energy density of 4970 Wh. Kg^{-1} and 3870 Wh. L^{-1} [14–21]. However, the lithium hydride has some challenges including its high formation energy (−90.5 kJ/mol) and high desorption temperature (700–1000 K) [20–24]. In addition, the lithium hydride reacts easily with humidity to irreversibly generate surface corrosion layers such as lithium oxide and lithium hydroxide. However, the controlled hydrolysis of LiH under low relative humidity is considered as an efficient hydrogen generator for mobility applications [25–27].

There are some attempts to overcome the thermodynamic problems of LiH. For example, it was shown using mechanical milling that the thermodynamic stability of the mixture between the lithium hydride and the silicon can be decreased with storing 5 wt% hydrogen at 490 °C [28]. In similar work, the mixture between the lithium hydride and the germanium can also decrease the stability of the system and store 3 wt% hydrogen at 400 °C [29,30]. In other work, the carbon is also one of the additives which can be improved the absorption/desorption of H_2 by decreasing the temperature below 200 °C [31]. Lei Wang and his co-authors [20] synthesized LiH nanoparticles via the catalytic hydrogenation of Li and showed that these nanoparticles once coated with Ni can release hydrogen from 100 °C. Pluengphon et al. [32] examined the

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Na–Li–H compositions by mixing cations of Na and Li under high pressure, utilizing ab initio random structure searching and cluster expansion techniques. The results showed that high pressure promotes the structural stability of Na_3LiH_{12}, $Na_2Li_2H_{12}$, and $NaLi_3H_{12}$ compounds, which was confirmed by phonon and elastic constant calculations.

A novel method is proposed in this work to improve the stability and decomposition temperature of the hydride LiH to be useful for different applications of hydrogen without reducing its high gravimetric density. The study aims to determine how Li vacancies affect the properties of hydrogen storage and to provide a reasonable justification for the observed reduction using DOS.

2. Calculation method

Throughout this paper, the AkaiKKR program package is chosen to perform our calculations because it allows us to control different concentrations of lithium vacancies in the material by using the coherent potential approximation (CPA) method combined in the package [33]. The generalized gradient approximation and the local density approximation (GGA/LDA) are used in calculating the density of states and total energy of $Li_{1-x}H$ and Li_{1-x}, respectively, where x is the concentration of Li vacancies inside the material [34,35]. A k-point grid of $10 \times 10 \times 10$ is used in this calculation to achieve more accurate results. The self-consistent field method is applied to describe the system approximately with a self-consistent convergence criterion of 10^{-6} Ry. The angular momentum is fixed at $l = 2$ for Li and LiH with or without vacancies.

The lithium hydride (LiH) has a NaCl-type structure (space group N° 225, Fm-3m) with unit cell parameters $a = b = c = 4.083$ Å. The atomic position for lithium atoms is (0, 0, 0) and for hydrogen atoms is (0.5, 0.5, 0.5) [36]. The lithium element (Li) has a body-centered cubic structure with a space group N° 229 (Im-3m symmetry) and unit cell parameters $a = b = c = 3.510$ Å. The atomic position for lithium atoms is (0, 0, 0) [37] (Fig. 1).

3. Results and discussion

3.1. Structural properties

The energy of formation and the temperature of desorption of the studied hydrides will be calculated using the total energy. Therefore, to get more precise results in the total energy, the cell parameters for both lithium hydride (LiH) and the pure element Li are obtained by using the relaxation method.

The experimental values ($a = 4.083$ Å) [36] and ($a = 3.510$ Å) [37] of LiH and Li, respectively, are initially used as the lattice constant for the relaxation method. The optimization of LiH and Li is evaluated by calculating the total energy for each variation in the cell volume as shown in Fig. 2.

The same process is used in order to optimize the lattice parameters and calculate the minimal total energies for both LiH and Li with different concentrations of vacancies.

The optimized parameters 'a' for both LiH and Li are 4.129 Å and 3.439 Å, respectively. These results are consistent with previously obtained values for LiH ($a = 4.083$ Å) [36] and for Li ($a = 3.510$ Å) [37]. This concordance with the previous results allows us to validate our calculation methods, which mainly include the use of GGA approximation for LiH and LDA approximation for Li.

The lattice parameters of highly vacant LiH and Li structures can be significantly affected by the distortion. The lattice parameters determine the size and shape of the unit cell, and any change in the lattice parameters can affect the physical and chemical properties of the material. Therefore, the creation of vacancies leads to reduce the lattice parameters of LiH structure. Regarding stability, the presence of vacancies makes the structure less stable as the vacancies can serve as nucleation

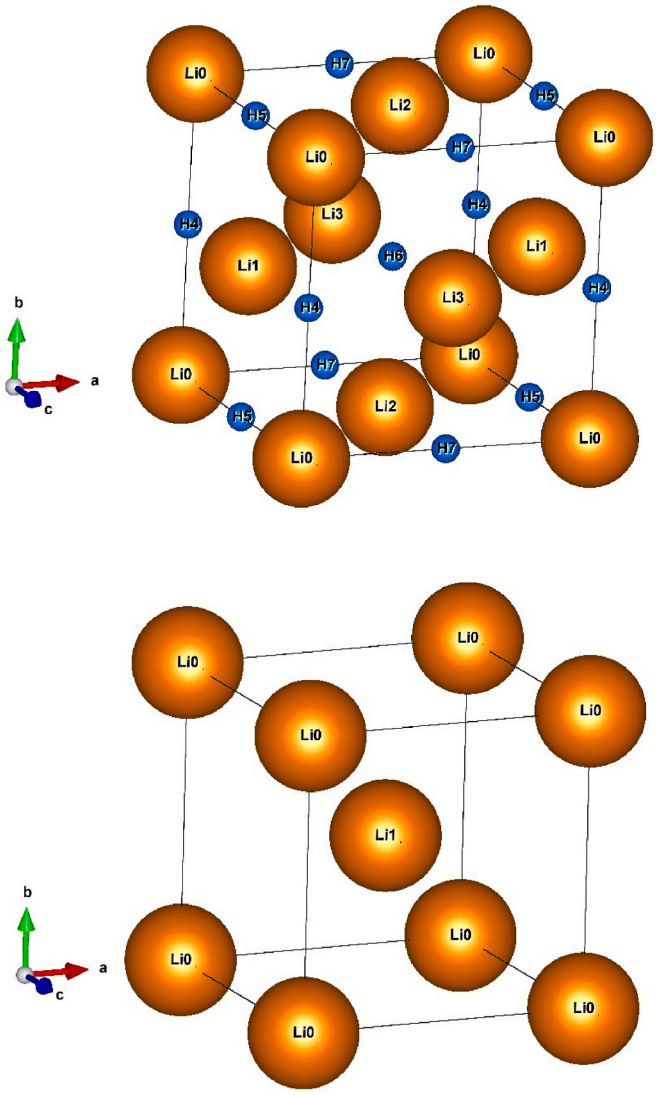


Fig. 1. Unit cell of LiH and pure element Li.

sites for crack formation, leading to fracture and failure of the material. Therefore, the LiH and Li structures have relaxed during the creation of vacancies for different concentrations. This was done to ensure that the resulting structure is energetically stable and can provide accurate and reliable results.

3.2. Energy of formation

Two thermodynamic parameters are required to analyze the stability of the hydride LiH with or without vacancies, namely the energy of formation (ΔH) and the temperature of desorption (Tdes). This part will focus only on calculating the formation of energy of the studied systems, and the findings will be used to determine the temperature of desorption in the next section.

The formation of the pure LiH is done according to the following reaction:

$$2Li + H_2 \rightarrow 2LiH \quad (1)$$

Using the difference between the total energy of products and reactants, the formation energy of our studied hydrides can be calculated by Ref. [38]:

$$\Delta H = \sum E_{tot}(\text{products}) - \sum E_{tot}(\text{reactants}) \quad (2)$$

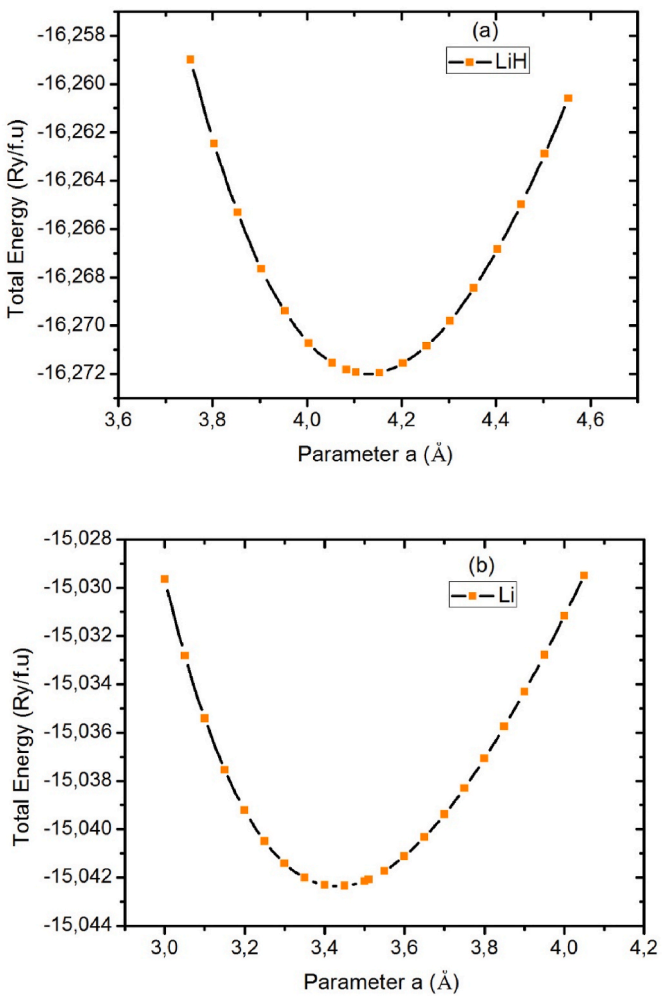


Fig. 2. Total energy as a function of parameter a for LiH (a) and Li (b).

By projecting the relation (2) on the reaction (1), the formation energy of LiH is written as follows:

$$\Delta H(LiH) = 2E_{tot}(LiH)_{relax} - 2E_{tot}(Li)_{relax} - E_{tot}(H_2) \quad (3)$$

where $E_{tot}(LiH)_{relax}$, $E_{tot}(Li)_{relax}$, and $E_{tot}(H_2)$ are the total energy of the lithium hydride, lithium element, and hydrogen molecule, respectively. The AkaiKKR package is utilized to calculate the total energies through electronic structure calculations based on the KKR method. To calculate the total energy of the studied systems, the initial step involves preparing the input file that contain information regarding the crystal structure, lattice parameters, atomic positions, and choice of exchange-correlation functional. Subsequently, the AkaiKKR calculation is executed, which involves solving the Kohn-Sham equations. Finally, the total energy value of the studied system is extracted from the output files. To obtain a more accurate total energy value for the studied systems, the relaxation method is employed in addition to the aforementioned steps. It helps ensure that the calculated total energy is at the minimum energy state for the system, providing a more accurate representation of the true energy of the studied system. In addition, the total energy of hydrogen is taken from Ref. [39].

The same approach is used for calculating the formation energy of the lithium hydride containing vacancies $Li_{1-x}H$ ($0 \leq x \leq 0.25$):

$$2Li_{1-x} + H_2 \rightarrow 2Li_{1-x}H \quad (4)$$

$$\Delta H(Li_{1-x}H) = 2E_{tot}(Li_{1-x}H)_{relax} - 2E_{tot}(Li_{1-x})_{relax} - E_{tot}(H_2) \quad (5)$$

where $E_{tot}(Li_{1-x}H)_{relax}$ and $E_{tot}(Li_{1-x})_{relax}$ are the total energy of the lithium hydride containing vacancies and the lithium element containing vacancies, respectively. The AkaiKKR package is used to calculate the total energy of LiH and Li with vacancy concentrations ranging from 1 to 25%, The calculation involves several steps. Firstly, an input file is prepared for LiH and Li with the desired concentration of vacancies. Next, the AkaiKKR program is executed to perform a self-consistent calculation of the electronic structure for the given system. Thirdly, the total energy is extracted from the output file. To obtain the total energy as a function of vacancy concentrations, this process is repeated for different vacancy concentrations ranging from 1 to 25%. The relaxation method is used alongside other steps to improve the results. In Table 1, there is a list of some calculated values of the total energy and the energy of formation.

Based on our calculations, LiH has a high stability and that is due to its formation energy which is equal to −91.371 kJ/mol. H_2. This value is in perfect agreement with that measured experimentally −90.5 kJ/mol. H_2 [22] or obtained theoretically −87 kJ/mol. H_2 [22].

The high thermodynamic stability of lithium hydride prevents its application for mobile hydrogen storage. So, as an attempt to reduce this stability, the effect of Li vacancies ($Li_{1-x}H$) can be a good approach to reach this goal. Thus, in this work, the concentration of lithium vacancies will be varied from 1 to 25% (until the value of the formation energy becomes positive). The different results are plotted in Fig. 3.

As shown in Table 1, the formation energy of LiH is increased with the concentration of lithium vacancies from −91.371 for x = 0 to −3.021 kJ/mol. H_2 for a value of 25% as defect concentration. This result also allows us to improve the temperature of desorption in LiH, as we will see in the next section. The explanation of this trend will be discussed using the density of states in Electronic Structure section.

3.3. Temperature of desorption

The second parameter in our thermodynamic properties is the desorption temperature (Tdes). Based on the standard Gibbs free energy ΔG, the temperature of desorption of the hydrides $Li_{1-x}H$ with $0 \leq x \leq 0.25$ can be calculated by:

$$\Delta G = \Delta H - T\Delta S \quad (6)$$

where ΔS and ΔH are the entropy and the energy of formation change of the reaction of decomposition, respectively.

In equilibrium, the decomposition temperature of the hydride can be estimated by the following expression:

$$T_{des} = -\frac{\Delta H}{\Delta S} \quad (7)$$

where the hydride entropy changes during dehydrogenation is related to the evolution of the hydrogen gas with a value of: ΔS = −130.7 J/Kmol. H_2 [40,41]. It is important to note that the calculation of entropy change due to hydrogen gas production is dependent on several factors such as temperature, pressure, and the specific metal hydride under study. It is necessary to consider the thermodynamic data and conditions used to calculate the entropy change and ensure their applicability to the specific system under study. The results are listed in Table 1 and illustrated in Fig. 3.

Based on our calculations, the T_{des} of LiH is ∼ 700 K. This value is very high and in agreement with that announced in Ref. [23]. In addition, Fig. 3 clearly shows that when the concentration increases, the temperature of desorption decreases from 699.09 (x = 0%) to 23.12 K (x = 25%). As a result, it is possible to control the temperature of desorption by varying the concentrations of lithium vacancies to reach the optimal range of the proton-exchange membrane fuel cell (PEMFC) [42]. This means that the ideal range for vacancy concentrations is estimated to be between 11.4% and 15.2%. Any concentration exceeding 15.2% is not advantageous as it would result in a

Table 1
Energy of formation, temperature of desorption and total energy of $Li_{1-x}H$ and Li_{1-x}.

Concentration of vacancies (x)	$Li_{1-x}H$		Li_{1-x}		Energy of formation (KJ/mol. H_2)	Temperature of desorption (K)
	System	Total energy (Ry/f.u)	System	Total energy (Ry/f.u)		
0	LiH	−16.27196	Li	−15.04235	−91.371	699.09
0.01	$Li_{0.99}H$	−16.11630	$Li_{0.99}$	−14.89021	−86.757	663.79
0.03	$Li_{0.97}H$	−15.80694	$Li_{0.97}$	−14.58601	−79.982	611.95
0.05	$Li_{0.95}H$	−15.49772	$Li_{0.95}$	−14.28193	−73.245	560.41
0.07	$Li_{0.93}H$	−15.18859	$Li_{0.93}$	−13.97797	−66.374	507.83
0.09	$Li_{0.91}H$	−14.87957	$Li_{0.91}$	−13.67417	−59.457	454.91
0.11	$Li_{0.89}H$	−14.57064	$Li_{0.89}$	−13.37049	−52.714	403.32
0.13	$Li_{0.87}H$	−14.26177	$Li_{0.87}$	−13.06693	−45.726	349.85
0.15	$Li_{0.85}H$	−13.95300	$Li_{0.85}$	−12.76351	−38.727	296.31
0.17	$Li_{0.83}H$	−13.64437	$Li_{0.83}$	−12.46023	−31.683	242.41
0.19	$Li_{0.81}H$	−13.33581	$Li_{0.81}$	−12.15707	−24.598	188.20
0.21	$Li_{0.79}H$	−13.02733	$Li_{0.79}$	−11.85406	−17.420	133.28
0.23	$Li_{0.77}H$	−12.71898	$Li_{0.77}$	−11.55118	−10.229	78.27
0.25	$Li_{0.75}H$	−12.41073	$Li_{0.75}$	−11.24843	−3.021	23.12

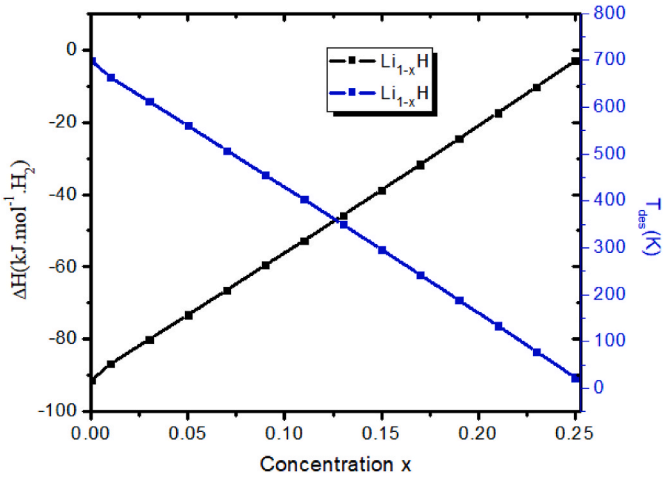


Fig. 3. Energy of formation and temperature of desorption as a function of the defect concentration x in $Li_{1-x}H$.

decomposition temperature that falls outside the optimal range for practical hydrogen storage applications in PEMFCs.

3.4. Gravimetric density

The gravimetric density of LiH has to be influenced by creating vacancies inside the material. Therefore, the following expression is used to study the effect of the concentrations of lithium vacancies on the gravimetric density of LiH:

$$D_g = \frac{m(H)}{(1-x) * m(Li) + m(H)} \quad (8)$$

where m(H) = 1.00784 u and m(Li) = 6.941 u are the atomic masses of hydrogen and lithium, respectively.

We can see that the gravimetric density of LiH is increased with the increase of the concentrations of lithium vacancies from 12.679 wt% (x = 0%) to 14.620 wt% for x = 15.2%.

In summary, the performed study shows that the hydrogen storage properties of LiH is improved by creating Li vacancies in the hydride and that can be seen by increasing both its formation energy and gravimetric capacity and decreasing its desorption temperature.

3.5. Electronic structure

The electronic structure in LiH can be used to understand the effect of vacancies on the thermodynamic properties including the heat of formation and the temperature of decomposition. Using the GGA approximation and the relaxed parameters, the density of states of LiH with and without vacancies are shown in Fig. 4. The unit cell size of the studied systems is expressed in terms of the Bohr radius unit. Due to consistent findings across all computations of $Li_{1-x}H$ with $0 \leq x \leq 0.25$, only the density of states (DOS) of $Li_{0.99}H$, $Li_{0.90}H$ and $Li_{0.75}H$ are plotted.

Based on this figure, it is evident that there is two parts in the valence band (VB), the first one is formed from a strong hybridization between p (Li) and s(H) states and the second part is also composed of a high hybridization, but this time between Li-2s and H-1s states. The conduction band (CB) is mainly attributed to Li-2p state.

Based on Fig. 4, Firstly, for the pure LiH, the high stability and, consequently, the high decomposition temperature are related to the strong interaction between H and Li atoms. Secondly, the observed decrease in the stability as a function of the concentration can be explained by two reasons. First, there is a decrease in Li atoms that form strong bonds with hydrogen atoms. Second, there is a shift of the lithium and hydrogen states towards the conduction band. As the concentration of vacancies in the material increases from 0 to 16%, there is a more noticeable shift in the lithium and hydrogen states. Consequently, the stability decreases in LiH with improving in its thermodynamic properties. The shifting of electronic states of Li and H into the conduction band is a crucial factor that affects the stability of these systems. Specifically, when Li atoms are removed from the system, the remaining Li and H atoms have more room to move, which leads to the shifting of their electronic states into the conduction band. Moreover, the integration of Li and H electronic states up to the Fermi level plays a crucial role in determining the stability of the system. The number of electrons within each unit cell corresponds to the integrated electronic states, which means that any change in the number of Li atoms also affects the number of Li valence states. This change in the number of electrons affects the electronic structure of the system, which in turn affects its stability.

4. Conclusions

The improvement of the thermodynamic properties of LiH is one of the challenges that face this hydride to be used in several hydrogen storage applications. In this context, we have investigated vacancy defects to improve the temperature of desorption and the energy of formation in LiH. In particular, we have varied the concentration of lithium vacancies from 1 to 25% in the hydride, and we have observed that the thermodynamic characteristics have been enhanced by increasing the energy of formation and decreasing both the stability and the temperature of desorption in LiH. Additionally, we have also found that the gravimetric density of LiH increases from 12.679 to 14.620 wt% when the concentration of lithium vacancies increases from 0 to 16%. This improvement in LiH is clarified using DOS, which clearly shows that

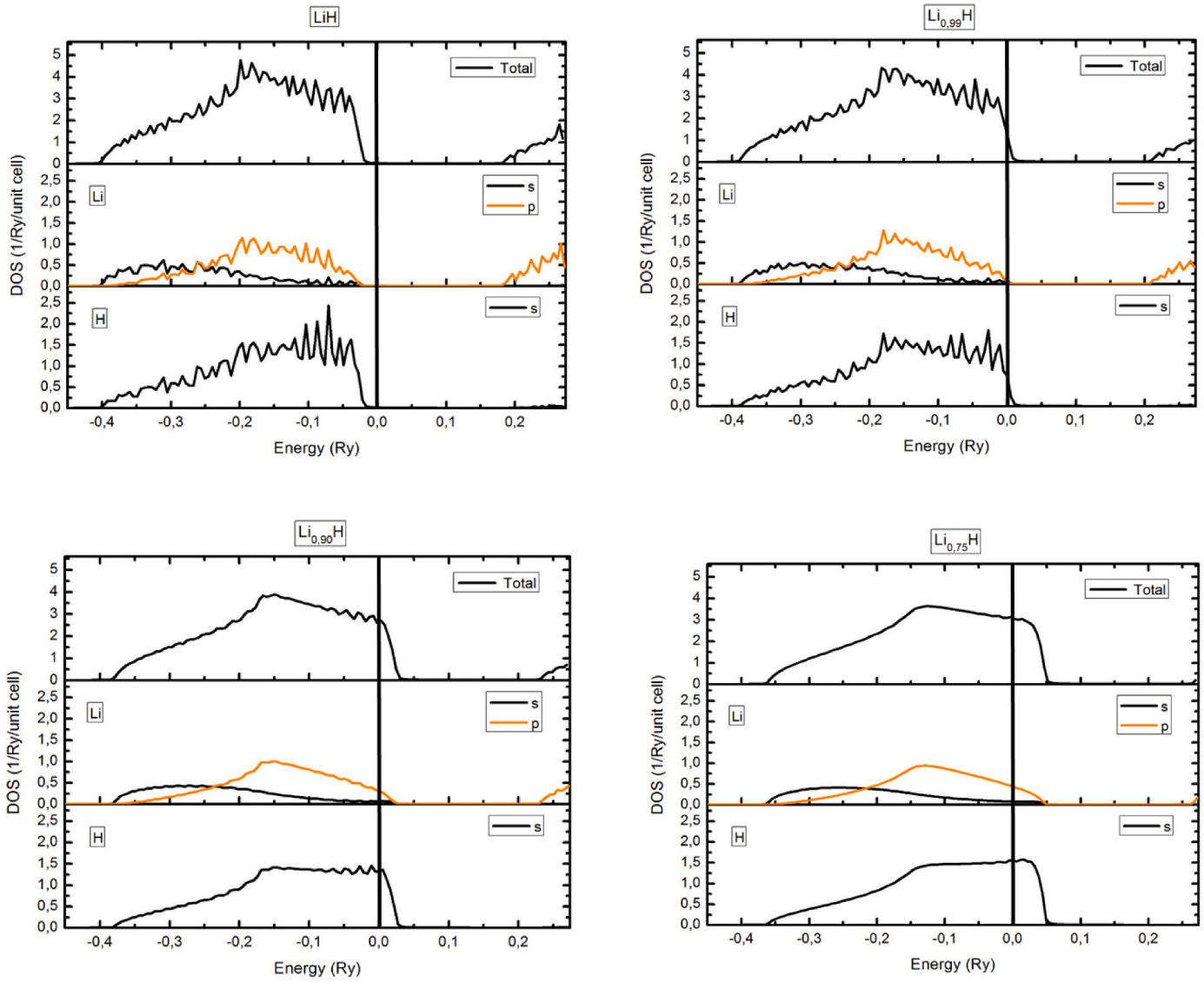


Fig. 4. Total and partial DOS of LiH, $Li_{0.99}H$, $Li_{0.90}H$ and $Li_{0.75}H$.

when the concentration of lithium vacancies increases in LiH, the lithium and hydrogen states shift towards the conduction band.

Author statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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